

UNIVERSITY OF TUNIS
TUNIS BUSINESS SCHOOL

DATA MINING PROJECT

Predicting Online Purchase Intention

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Academic Year: 2025–2026 | Spring Semester

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1. Dataset Description

1.1 General Context

This project uses the Online Shoppers Purchasing Intention dataset, which captures behavioral and session-level data of users visiting an e-commerce website. Each record corresponds to a unique browsing session and includes information about the pages visited, time spent on those pages, technical characteristics of the visit, and whether or not the session ended in a purchase.

The dataset is particularly relevant to e-commerce analytics, where understanding what drives a user to complete a purchase is a key business challenge. By analyzing session-level data, businesses can identify patterns in user behavior that distinguish buyers from non-buyers and use these insights to improve conversion rates.

1.2 Source

The dataset is publicly available on the UCI Machine Learning Repository and Kaggle under the name **Online Shoppers Purchasing Intention Dataset**. It was collected by researchers studying e-commerce user behavior and contains data from a real online retail platform.

1.3 Structure of the Dataset

The dataset originally contains 12,330 observations and 18 variables. After removing 125 duplicate records during preprocessing, 12,205 observations remained.

Each observation represents **one user session on the e-commerce website**. The 18 variables fall into four categories:

Category	Variables	Description
Page Interaction	Administrative, Administrative_Duration, Informational, Informational_Duration, ProductRelated, ProductRelated_Duration	Number of pages visited and time spent on each page type during the session
Session Quality	BounceRates, ExitRates, PageValues, SpecialDay	Metrics reflecting user engagement and proximity to special commercial days
Session Context	Month, OperatingSystems, Browser, Region, TrafficType, VisitorType, Weekend	Technical and temporal context of the visit
Target Variable	Revenue	Boolean: True if the session resulted in a purchase, False otherwise

1.4 Key Observations

- No missing values were found in any of the 18 columns, no imputation was required.
- The dataset contains 10,422 non-purchase sessions (84.5%) and 1,908 purchase sessions (15.5%), confirming a significant class imbalance.
- Numerical features such as Administrative_Duration and ProductRelated_Duration show high variance, with maximum values far above their means, suggesting the presence of heavy users.

2 Problem Statement

2.1 Problem Definition

In e-commerce, a large proportion of website visitors leave without making a purchase.

Understanding which sessions are likely to result in a purchase allows businesses to make smarter, data-driven decisions across multiple areas:

Real-Time Personalization: When a high-intent session is detected, the platform can dynamically trigger targeted interventions such as personalized discounts, free shipping offers, or limited-time promotions, far more effectively than applying blanket promotions to all visitors regardless of purchase probability.

Marketing Budget Allocation: A predictive model enables businesses to concentrate paid advertising, email campaigns, and retargeting budgets on high-conversion-probability users, maximizing return on investment across all acquisition channels.

Customer Service Prioritization: By flagging sessions with high purchase intent, businesses can proactively deploy live chat support toward users most likely to convert, optimizing resource allocation without increasing operational costs.

Inventory and Operational Planning: Behavioral patterns identified by the model such as which traffic sources or user segments convert at higher rates can inform demand forecasting, inventory stocking, and staffing decisions during high-conversion periods.

Website Optimization: Feature importance analysis reveals which page types and behavioral signals are most predictive of conversion, enabling data-driven UX improvements that reduce exit rates and increase overall conversion rates.

The central challenge is to build a classification model that can accurately predict, based on behavioral data collected during a browsing session, whether that session will result in a purchase or not.

2.2 Research Objective

The objective of this project is to apply supervised classification techniques to predict the binary target variable Revenue, which indicates whether a user session resulted in a completed purchase. Three models are trained, compared, and interpreted to identify the best performing approach and the most influential predictors of purchase intent.

2.3 Research Questions

- Can we accurately predict whether a user session will result in a purchase based on behavioral and session data?
- Which features are the most informative predictors of purchase intent?
- Which classification model performs best on this dataset, and why?

2.4 Target Variable

The target variable is Revenue (binary: 0 = no purchase, 1 = purchase). As shown in Figure 1, the dataset is heavily imbalanced, with 84.5% of sessions resulting in no purchase and only 15.5% resulting in a purchase. This imbalance is acknowledged as a key limitation and is discussed in the conclusion.

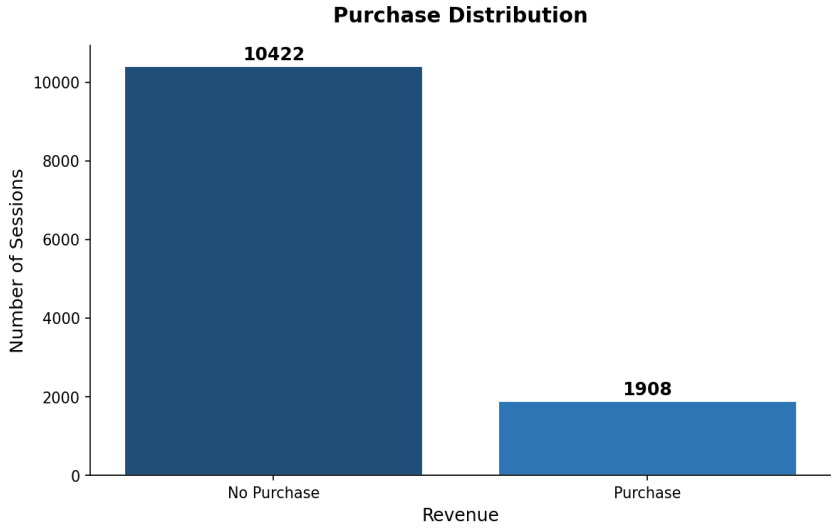


Figure 1: Distribution of the target variable Revenue across all sessions

3. Data Preparation

3.1 Missing Values

The first preprocessing step was to verify data completeness using **df.isnull().sum()**. The result confirmed that all 18 columns contain zero missing values across all 12,330 observations. No imputation was therefore required, and the dataset can be used directly for modeling after other preprocessing steps.

3.2 Removing Duplicate Records

Duplicate rows were removed using **df.drop_duplicates()**. A total of 125 duplicate records were identified and removed, reducing the dataset from 12,330 to 12,205 observations. Keeping duplicate records would bias the model by causing it to learn the same pattern multiple times, artificially increasing its confidence on those specific cases and reducing its ability to generalize.

3.3 Encoding Categorical Variables

Machine learning models cannot process text values directly. The categorical columns Month and VisitorType were converted into binary numerical columns using **pd.get_dummies()**. The parameter **drop_first=True** was used to remove the first category from each encoded group. This avoids the dummy variable trap, which occurs when one column can be perfectly predicted from the remaining encoded columns, introducing multicollinearity into the model and making it unstable.

The boolean columns Weekend and Revenue were converted from True/False to integer values 0 and 1 using **astype(int)**. After encoding, the feature matrix X contains 26 columns.

3.4 Train/Test Split with Stratification

The dataset was split into **80%** training data (9,764 observations) and **20%** testing data (2,441 observations) using **train_test_split from scikit-learn**. The parameter **random_state=42** ensures full reproducibility of results.

The parameter **stratify=y** was applied to ensure that both the training and testing sets contain the same proportion of buyers and non-buyers as the original dataset (84.5% / 15.5%). Without stratification, a purely random split could distribute the minority class unevenly between the two sets, making model evaluation unreliable. This is especially important given the class imbalance present in this dataset.

3.5 Feature Scaling

StandardScaler() was applied to generate scaled versions of the training and testing features, stored separately as **X_train_scaled** and **X_test_scaled**. This scaling was applied exclusively for use with **Logistic Regression**, which relies on gradient-based optimization and is sensitive to differences in feature magnitude , features with large numerical ranges would otherwise dominate the model.

Decision Tree and Random Forest are tree-based models that make predictions by applying threshold-based splits on individual features. Since the absolute scale of a feature does not affect where the tree places its splits, scaling has no effect on these models. The original unscaled data was therefore used for both tree-based methods.

The scaler was fitted only on the training data using **fit_transform(),** and only transform was applied to the test data. This is a critical step to prevent data leakage, which would occur if information from the test set were used to compute the scaling parameters.

4. Methods and Analysis

Three classification models were applied and compared in this project. Each model was selected for a specific reason, and the combination allows for a meaningful comparison between a linear model, a simple tree-based model, and an ensemble approach.

4.1 Logistic Regression

Logistic Regression was chosen as a baseline model for two specific reasons. First, the problem is a binary classification task a session either results in a purchase or it does not which is exactly the type of problem Logistic Regression is designed for. Second, it allows us to establish a performance benchmark against which the more complex models, Decision Tree and Random Forest, can be compared. If a simple linear model already performs well, there is less justification for using more complex approaches.

However, Logistic Regression assumes a linear relationship between the input features and the log-odds of the outcome. Given the behavioral nature of our dataset, where interactions between features such as PageValues, ExitRates, and session duration are likely non-linear, a linear model may not capture the full complexity of purchase intent. This is precisely why we complement it with tree-based models that can capture non-linear patterns.

The model was configured with **max_iter=1000** to ensure full convergence during training, and trained exclusively on the scaled feature set **X_train_scaled**, since Logistic Regression is sensitive to feature magnitude.

4.2 Decision Tree

Decision Tree was chosen for this project because it can naturally capture the non-linear relationships present in e-commerce behavioral data. Unlike Logistic Regression, it does not assume any linear relationship between features and the target variable, making it more suitable for datasets where purchase intent is driven by complex interactions between variables such as PageValues, BounceRates, and session duration.

It also serves an important role in this project as an interpretable middle ground between the simple linear baseline of Logistic Regression and the more complex ensemble approach of Random Forest, allowing us to assess whether a single tree structure is sufficient to capture the patterns in the data.

To prevent overfitting, **max_depth=10** was applied, limiting the tree to 10 levels and forcing it to learn general patterns rather than memorizing the training data. Since Decision Trees are threshold-based models and not sensitive to feature magnitude, the model was trained on unscaled data.

4.3 Random Forest

Random Forest was chosen as the most advanced model in this project because it directly addresses the main weakness of a single Decision Tree, which is its tendency to overfit. By building 200 independent trees, each trained on a different random subset of the data and features, and combining their predictions through majority vote, Random Forest produces more stable and accurate results than any individual tree could achieve alone.

It is particularly well suited for this dataset because e-commerce behavioral data tends to be noisy and high-dimensional, with complex interactions between features. The ensemble nature of Random Forest makes it robust to this noise and capable of capturing subtle patterns in purchase intent that a single tree or a linear model would miss.

Additionally, Random Forest provides feature importance scores, which is especially valuable for this project as it allows us to identify which behavioral variables are the strongest predictors of purchase intent, adding a layer of business interpretation beyond just model performance.

The model was configured with **n_estimators=200** for greater prediction stability, and **max_depth=10** was applied to each individual tree to control overfitting. Since Random Forest is a threshold-based model, it was trained on unscaled data.

5. Results and Interpretation

5.1 Model Performance

The three models were evaluated on the test set using four metrics: **Accuracy, Precision, Recall, and F1 Score**.

Since the dataset is **imbalanced**, F1 Score is **the primary comparison metric** as it accounts for both false positives and false negatives and is not inflated by the dominant majority class.

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	88.94%	77.18%	41.62%	0.541
Decision Tree	88.90%	65.46%	61.52%	0.634
Random Forest	90.25%	74.83%	56.81%	0.646

5.2 Interpretation of Results

Logistic Regression achieves the highest **precision** (77.18%), meaning that when it predicts a purchase, it is correct most of the time. However, its **recall** is only 41.62%, meaning it misses more than half of all actual purchases. The model is overly conservative, it avoids false alarms but at the cost of missing real buyers. This results in the lowest **F1 Score** of 0.541.

Decision Tree achieves the best **recall** (61.52%), catching the most actual buyers of the three models. However, its precision drops to **65.46%** due to a higher number of false alarms. The balance between the two metrics yields an F1 Score of **0.634**.

Random Forest delivers the strongest overall performance with the highest **accuracy** (90.25%) and the highest **F1 Score** (0.646). It maintains strong **precision** (74.83%) while achieving better **recall** than Logistic Regression (56.81%). **This balance makes it the most reliable and practically useful model for this classification problem.**

5.3 Confusion Matrix — Random Forest

The confusion matrix for the best performing model, Random Forest, is shown in Figure 2. It breaks predictions into four categories: **True Negatives** (correctly predicted non-purchases), **True Positives** (correctly predicted purchases), **False Positives** (predicted purchase but wrong), and **False Negatives** (missed actual purchases).

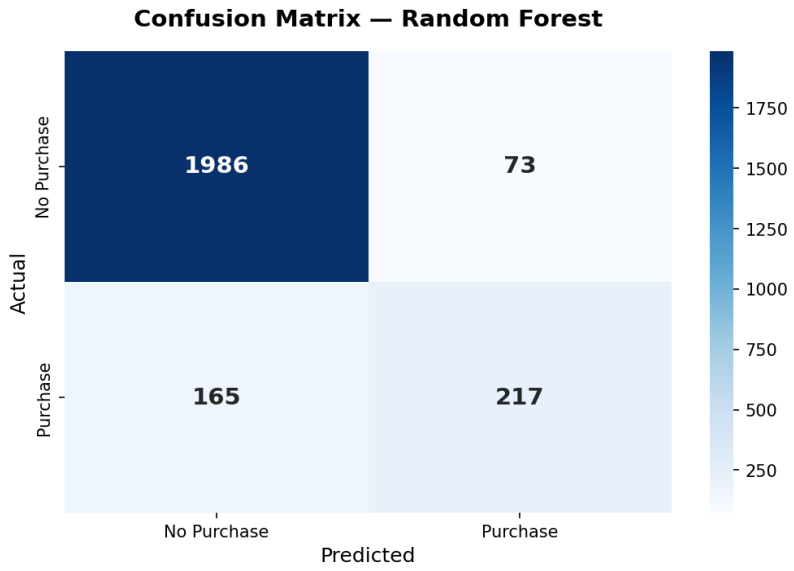


Figure 2: Confusion Matrix for Random Forest on the test set

	Predicted: No Purchase	**Predicted: Purchase**
Actual: No Purchase	1,986 (True Negatives)	73 (False Positives)
Actual: Purchase	165 (False Negatives)	217 (True Positives)

The model correctly identified 1,986 non-purchase sessions and 217 purchase sessions out of 2,441 total test observations. The very low number of false positives (73) confirms the high precision of Random Forest when it predicts a purchase, it is almost always correct, making it a reliable tool for triggering targeted business interventions without wasting resources on unlikely buyers.

The main weakness is the 165 false negatives, representing real purchase sessions the model failed to identify. This is a direct consequence of the class imbalance in the dataset , with only 15.5% of sessions resulting in a purchase, the model is naturally biased toward the majority class, causing some genuine buyers to be missed.

5.4 F1 Score Comparison

Figure 3 visually summarizes the **F1 Score of all three models** side by side, confirming that **Random Forest** outperforms both alternatives.

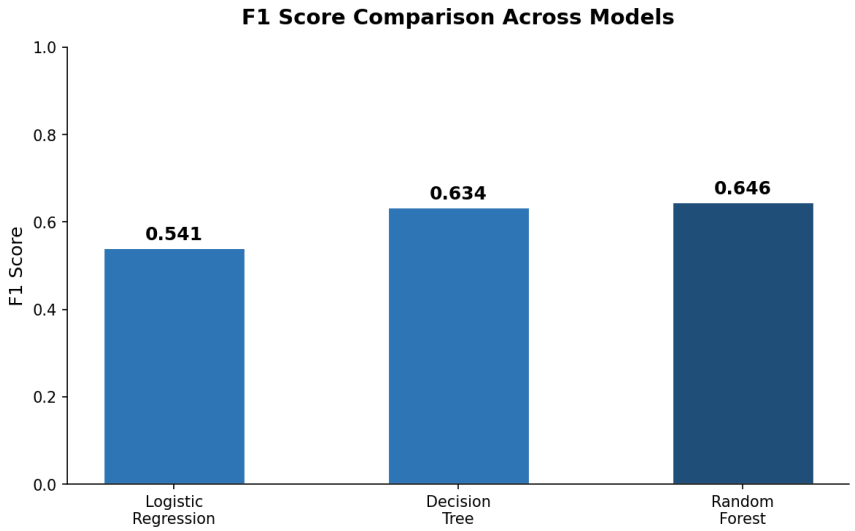


Figure 3: F1 Score comparison across Logistic Regression, Decision Tree, and Random Forest

5.5 Feature Importance — Random Forest

Figure 4 displays **the top 10 most important features** identified by the Random Forest model. Feature importance is calculated by measuring how much each feature reduces impurity (Gini impurity) on average across all 200 trees.

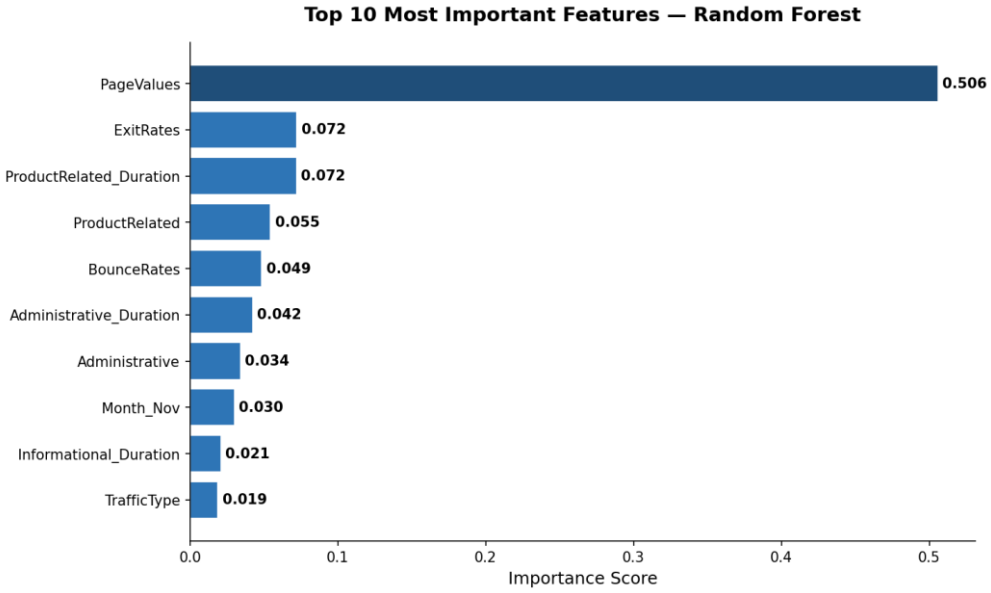


Figure 4: Top 10 most important features ranked by Random Forest importance score

	Feature	Score	Business Interpretation
1	PageValues	0.506	Users who visit high-value pages are far more likely to purchase. Businesses should prioritize driving traffic toward these pages and ensure they are optimized for conversion.
2	ExitRates	0.072	Pages with high exit rates are actively killing conversions. Businesses should identify these pages and redesign them to retain users longer.
3	ProductRelated_Duration	0.072	Users who spend more time on product pages are genuinely considering a purchase. Businesses can target these users with timely promotions to push them toward checkout.
4	ProductRelated	0.055	Users browsing multiple product pages show strong purchase intent. Retargeting these users with personalized recommendations can increase conversion rates.
5	BounceRates	0.049	Sessions with no interaction after landing. High bounce rates are a strong negative signal.
6	Administrative_Duration	0.042	Time spent on account or order pages suggests the user is actively managing a purchase. These sessions deserve priority customer support intervention.
7	Administrative	0.034	Users visiting multiple administrative pages may be experiencing friction in the checkout process. Simplifying account management can reduce drop-off.
8	Month_Nov	0.030	Conversion rates spike during November due to promotional periods (Black Friday). Businesses should prepare inventory, staffing, and campaigns well in advance of this window.
9	Informational_Duration	0.021	Users spending time on informational pages are still in the research phase. Businesses can use this signal to serve educational content that nudges them toward a decision.
10	TrafficType	0.019	Certain acquisition channels consistently convert better than others. Businesses should reallocate marketing spend toward the highest-converting traffic sources.

6. Conclusion

6.1 Summary

This project applied **three supervised classification models** to the Online Shoppers Purchasing Intention dataset with the objective of **predicting whether a user browsing session would result in a purchase**. The dataset was prepared through duplicate removal, categorical encoding, stratified train/test splitting, and selective feature scaling. Three models: **Logistic Regression**, **Decision Tree**, and **Random Forest** were trained and evaluated using **Accuracy**, **Precision**, **Recall**, and **F1 Score**.

6.2 Main Findings

Random Forest was identified as the best performing model, achieving an **accuracy** of **90.25%** and an **F1 Score** of **0.646**. Its ensemble architecture, which combines **200** independently trained decision trees, produces more stable and accurate predictions than either a single tree or a linear model. Feature importance analysis revealed that **PageValues** is by far the most influential predictor, accounting for approximately half of all predictive power in the model. **ExitRates**, **ProductRelated_Duration**, and **ProductRelated** are the next most important features.

From a business perspective, the key insight is that the monetary value associated with the pages a user visits is a much stronger predictor of purchase intent than the duration or breadth of their browsing.

E-commerce platforms should prioritize tracking and optimizing high-value page interactions, as these represent the clearest signals that a session is likely to convert. Campaigns targeting users who display high PageValues behavior are more likely to yield conversions.

6.3 Limitations

The primary limitation of this analysis is the **class imbalance** in the dataset. With **84.5%** of sessions resulting in no purchase, models are **naturally biased** toward predicting **the majority class**. This is reflected in the relatively **high number** of **false negatives** across **all models**, particularly in Random Forest (165 missed purchases). Techniques such as oversampling with SMOTE, undersampling, or applying class_weight parameters could be explored in future work to improve recall and reduce missed purchases.

Additionally, no systematic hyperparameter optimization was performed. A grid search approach using GridSearchCV could further improve model performance. Cross-validation was also not applied, meaning evaluation is based on a single train/test split, which introduces some variability in the reported metrics. Future work could address these limitations to produce more robust results.